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WHAT IS THE CLASSICAL REGGE LIMIT FOR $n \geq 5$ PARTICLES? AN OPEN PROBLEM, WITH COMPUTATIONAL EVIDENCE FROM A FIVE-PHOTON DEGREE-11 SEARCH

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ABSTRACT. The Classical Regge Growth (CRG) conjecture — that any consistent classical S -matrix grows no faster than s^2 in the Regge limit $s \rightarrow \infty$ at fixed t — is well-posed and has been used to constrain four-particle photon and graviton S -matrices [1]. Extending this to $n \geq 5$ particles requires a generalization of “the Regge limit” itself, and we show this generalization is genuinely ambiguous: no definition exists in the literature, and the closest established framework with a similar name — multi-Regge kinematics, from the gauge-theory literature on multi-particle production [3] — is motivated by different physics and is not obviously the correct generalization of the causality-derived CRG bound. Using the degree-9 and degree-11 five-photon operator basis constructed in a companion note [4], we construct three qualitatively distinct, explicitly validated families of $n = 5$ Regge trajectories and search for operators satisfying the conjectured bound. We find: the unique degree-9 invariant has an exact growth floor of s^3 within one family; a 9-dimensional subspace of the (at-the-time) 15-dimensional degree-11 space satisfies the bound in that same family; a 2-dimensional subspace survives two qualitatively different families and two independent parameter configurations each; but a third family kills this subspace entirely, and an exact computation of the combined kernel across all three families, on the complete 16-dimensional degree-11 space (including a topology discovered after the two-family result), gives exactly $\{0\}$. We interpret this pattern — four structurally different attempts to enlarge the search all failing in the same way — as evidence that the obstruction is in the definition of the limit, not in an incomplete search, and we state this as the open problem.

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1. INTRODUCTION

The Classical Regge Growth (CRG) conjecture states that the tree-level S -matrix of any consistent classical theory grows no faster than s^2 in the Regge limit ($s \rightarrow \infty$ at fixed t) [1]. There is evidence for this from causality: the chaos bound in AdS/CFT, applied to the boundary dual of a bulk theory of scalars, gauge fields, or gravitons, implies CRG for the corresponding flat-space S -matrices [2]. Deriving CRG this way is substantial holographic technical work, well beyond the scope of this project; instead, following Chowdhury, Gadde, Gopakka, Halder, Janagal & Minwalla (CGGHJM) [1], we take CRG as a given conjecture and ask what it implies for the operator classification at $n = 5$, building on the representation-theoretic and combinatorial foundation constructed in a companion note [4].

At $n = 4$, “the Regge limit” has an unambiguous meaning: elastic $2 \rightarrow 2$ scattering at fixed momentum transfer t , realized physically by boosting all four momenta while correlating the scattering angle to keep t fixed as $s \rightarrow \infty$. At $n \geq 5$, there is no single kinematic invariant playing the role of “ t ”, and it is not obvious a priori which invariants should be held fixed while which others diverge. This note’s central finding is that this ambiguity is not merely bookkeeping: it appears to be the actual obstruction to extending the $n = 4$ classification program to $n = 5$, at least within the family of constructions we tried.

Remark 1.1 (Literature search). *A direct search of the arXiv API (both for “classical Regge growth” specifically, and for follow-up work on the $n = 4$ classification) turned up no paper addressing CRG at $n \geq 5$; the follow-up literature since 2019 extends the $n = 4$ story (massive spinning particles, dimensional reduction, the holographic chaos-bound derivation itself) but does not touch $n \geq 5$. Separately, there is an established, decades-old, and extensively used framework called multi-Regge kinematics (MRK) for n -point amplitudes at $n \geq 5$, going back to Del Duca [3] and used throughout the BFKL/planar- $\mathcal{N} = 4$ literature up to $n = 7$ -point amplitudes: particles are ordered by rapidity, and the limit takes all consecutive rapidity gaps to infinity (“strong rapidity ordering”) with transverse momenta fixed. We do not adopt MRK here, for a specific reason: MRK’s physical motivation is multi-particle production at high energy in gauge theory (an inherently different regime from the causality/chaos bound that motivates CRG), and nothing we are aware of establishes that MRK is the kinematic regime in which a chaos-bound-type causality argument would actually constrain growth to s^2 . Adopting it without that justification would repeat the same error as inventing an ad hoc generalization, only with a citation*

that lends false confidence. We regard this as itself part of the open problem, not a solved input.

Organization. Section 2 reviews the $n = 4$ Regge limit at the level of explicit momenta, validated as a template. Section 3 shows a naive generalization to $n = 5$ fails structurally. Section 4 constructs three qualitatively distinct families of $n = 5$ Regge trajectories. Section 5 reports the growth rates found at degrees 9 and 11, culminating in the negative result on the full 16-dimensional degree-11 space. Section 6 summarizes the verification methodology, including several bugs found and fixed. Section 7 discusses what this means and states the open problem.

2. THE $n = 4$ REGGE LIMIT, VALIDATED AT THE LEVEL OF EXPLICIT MOMENTA

We work in $D = 6$ throughout (consistent with [4]), with the standard realization of elastic $2 \rightarrow 2$ scattering at fixed t :

$$p_1 = -E(1, n_1), \quad p_2 = -E(1, -n_1), \quad p_3 = E(1, n_3), \quad p_4 = E(1, -n_3),$$

with n_1, n_3 unit spatial vectors at angle θ , giving $s = 4E^2 \rightarrow \infty$ and $t = -2E^2(1 - \cos \theta)$. Choosing $\cos \theta(E) = 1 + t_0/(2E^2)$ keeps $t = t_0$ exactly fixed as $E \rightarrow \infty$ ($\theta \rightarrow 0$, correlated with $E \rightarrow \infty$) — the standard forward-limit realization of “ $s \rightarrow \infty$ at fixed t ”. Crucially, all four momenta participate in the high-energy limit; no particle is held at rest.

Remark 2.1 (A gauge-reference bug, instructive for $n = 5$). *Constructing the polarizations ε_i requires a reference momentum to fix the residual gauge freedom (projecting a seed vector to satisfy $\varepsilon_i \cdot p_i = 0$). An initial attempt used a “ t -channel partner” as that reference (e.g. p_3 for ε_1) and found $(F^1:F^2)(F^3:F^4)$ growing as Λ^5 , not the Λ^4 expected by naive degree counting. Diagnosis: $\eta(p_1, p_3) = -t/2$ is held fixed by construction (it is literally the momentum transfer being kept constant), so using p_3 as a gauge reference divides by a quantity that does not grow, causing the transverse part of ε_1 to blow up spuriously. Using an s -channel partner instead (e.g. p_2 , with $\eta(p_1, p_2) = -s/2 \sim E^2$) fixes this: after the correction, $F^1:F^2 \sim s^{1.0000}$ and $(F^1:F^2)(F^3:F^4) \sim s^{2.0000}$, matching naive counting and CGGHJ’s own table (2-derivative structures grow as s , 4-derivative as s^2). This is the first instance, in either this note or [4], of a recurring lesson: in any Regge trajectory, using a “fixed-channel” pair as a gauge reference produces spurious growth unrelated to the physical S -matrix, and one must classify growing vs. fixed channels before constructing explicit polarizations.*

3. A NAIVE GENERALIZATION TO $n = 5$ FAILS STRUCTURALLY

The obvious generalization — boost two particles (p_1, p_2) along opposite null reference directions while holding three “spectators” (p_3, p_4, p_5) fixed — fails for structural reasons, not a computational obstacle. There are two related problems, the second a sharper version of the same underlying obstruction:

- (1) By momentum conservation, if $p_1 + p_2$ grows, at least one spectator must absorb the recoil and grow as well; one cannot hold $p_3 + p_4 + p_5$ fixed while $p_1 + p_2$ diverges, since their sum must vanish exactly.
- (2) No generic null momentum can have zero overlap with *both* reference null directions simultaneously: a null vector with vanishing components along

two independent null directions would live entirely in their spacelike transverse complement, where the only null vector is zero. Hence invariants like s_{13} , which one would naively want to hold fixed (as an analogue of t), necessarily grow too if p_3 is a generic null momentum.

At $n = 4$, the Regge limit works precisely because all four momenta participate in a specific, correlated boost-plus-forward-limit structure, not because two particles move and two stay put. Generalizing this correctly to $n = 5$ (where “holding t fixed” must mean something precise with five momenta) is a genuine modeling choice, not resolved in the literature for $n \geq 5$, and there is no evident reason to expect a unique correct choice.

4. THREE FAMILIES OF $n = 5$ REGGE TRAJECTORIES

We constructed three qualitatively distinct families, each validated independently (massless momenta, exact momentum conservation, and an explicit classification of which of the ten pairwise invariants grow vs. stay fixed, checked *before* trusting any polarization construction, per Remark 2.1).

4.1. Family 1: hierarchical hard-trio / recoil-pair. A “hard” cluster $\{1, 2, 3\}$ with $s_{12} \rightarrow \infty$, and a recoil pair $\{4, 5\}$ boosted from a two-body decay of a system with momentum $P_{45} = -(p_1 + p_2 + p_3)$. The energy E_3 and angle $\theta(E)$ of p_3 relative to p_1 are tuned (via the same forward-limit mechanism as Section 2) to hold *both* $t_{13} := -(p_1 + p_3)^2$ and the recoil invariant mass $M_{45}^2 := -(p_1 + p_2 + p_3)^2$ fixed as $E \rightarrow \infty$. The direction of the $\{4, 5\} \rightarrow$ two-body decay in the P_{45} rest frame cannot be generic (oscillating, no clean power law) or exactly aligned with the boost axis (degenerate); a small fixed tilt angle φ_0 gives clean growth in all ten pairs. Resulting channel classification: growing $(1, 2), (1, 4), (1, 5), (2, 3), (3, 4), (3, 5)$ (all exactly $\sim E^2$); fixed $(1, 3), (2, 4), (2, 5), (4, 5)$ (exact constants).

4.2. Family 2: two-scale hierarchical. Identical construction to Family 1, except the recoil invariant mass is no longer held fixed but grows at the same rate as the hard channel, $M_{45}^2 = m_0^2/x^2$ (writing $x = 1/E$). The algebraic derivation of $\cos\theta, E_3$ that fixes t_{13} and M_{45}^2 simultaneously is valid for *any* value of M_{45}^2 , including one depending on x , so no new derivation was needed. Resulting channel classification is qualitatively different: only $(1, 3)$ (i.e. t_{13}) stays fixed; all other nine pairs, including three that were fixed in Family 1, now grow as $\sim E^2$.

Remark 4.1 (Relabelings are not new families). *Every operator considered here is fully S_5 -symmetrized (summed over all 120 permutations, as in [4]). Consequently, evaluating such an operator on a trajectory built by relabeling which trio is “hard” and which pair is “recoil” gives numerically identical results to the original labeling — it is not an independent check. A genuine second family requires a qualitatively different growing/fixed pattern, not merely a different assignment of particle labels to the same pattern; we verified this obstruction by direct analysis before building Family 2, and it is the reason Family 2 changes the scaling of M_{45}^2 rather than which particles are “hard”.*

4.3. Family 3: fast recoil growth. Same construction again, with $M_{45}^2 \sim m_0^4/x^4$ (growing *faster* than the hard channel $s_{12} \sim 1/x^2$) — two powers of x^{-1} beyond Family 2. (A first attempt to instead scale t_{13} itself at an intermediate

rate broke the parity of the kinematics, giving half-integer powers of x incompatible with our exact-arithmetic machinery; this is consistent with the parity theorem of [4] and was diagnosed before any computation was attempted.) Channel classification: $(1, 3)$ still fixed; $(1, 2), (1, 4), (1, 5)$ grow as before ($\sim E^2$); but $(2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)$ now grow as $\sim E^4$ — *faster* than the nominally “hard” channel itself.

5. RESULTS

All computations use exact rational arithmetic (momenta and polarizations constructed in the quadratic extension $\mathbb{Q}(x)(S)$, $S^2 = R(x)$ a rational function — the only irrationality in the whole construction, since S always appears linearly until squared — followed by an exact Laurent series expansion in $x = 1/E$; see [4]’s methodology and Section 6 below for the fast backend used). Growth rates are read off as the lowest surviving power of x in this expansion, i.e. x^{-k} corresponds to growth E^k .

5.1. Degree 9. Within Family 1, the unique degree-9 invariant (within the topologies known before the sandwich atom was added) grows as $E^8 = s^4$, robust across four independent parameter configurations (exponent 4.00 ± 0.01 throughout). Once the sandwich atom M_{ab}^{ij} is included, the true dimension of the degree-9 space is 3 (not 1), and an *exact* rank/kernel computation on the Laurent coefficients (not a numerical extrapolation) proves that s^3 is the genuine floor of the *entire* 3-dimensional space in this trajectory: no combination can do better, confirmed identically in two independent parameter configurations.

5.2. Degree 11, Family 1 alone. Within the (pre-sandwich) 3-dimensional space, an analogous exact argument shows s^4 is the floor: any combination with a component along the two leading directions leaves an E^9 residue that the third cannot cancel. Once the sandwich atom is included systematically, the dimension of degree 11 jumps to (at least) 15 (documented fully in [4], whose exhaustive combinatorial sweep later confirmed this is exact within the four known atom types). An exact rank/kernel computation on this enlarged space reveals a **9-dimensional subspace** whose growth is $\leq E^4 = s^2$ in Family 1 — i.e. compatible with the conjectured CRG bound — identically reproduced in a second independent parameter configuration.

5.3. Degree 11, two families combined. The 9-dimensional Family-1 subspace does *not* survive Family 2: all nine directions fall to x^{-11} ($s^{5.5}$), violating the bound by a wide margin. However, computing the combined kernel of both families’ constraints directly (stacking the Laurent-coefficient rows of both trajectories and taking the exact nullspace) reveals a genuinely new, smaller **2-dimensional subspace** U_1, U_2 whose growth is $\leq s^2$ in *both* Family 1 and Family 2 simultaneously — verified by direct reconstruction (not just rank counting) and reproduced identically across two independent parameter configurations in each family. This is the first candidate in this project to survive more than one qualitatively distinct Regge trajectory.

5.4. Degree 11, three families combined: a null result. Family 3 kills U_1, U_2 outright: both fall to x^{-22} , the same generic order as any individual, unprotected basis element, violating the bound enormously. A single combination of U_1, U_2 can cancel that leading term, but would then need to independently satisfy roughly 18 more order-by-order constraints using zero remaining degrees of freedom — there is no structural reason to expect this, and we did not pursue the (rapidly growing) computational cost of checking it exhaustively.

Subsequently, a genuinely new atom (the “triple field-strength sandwich” T) was discovered, raising the degree-11 dimension from 15 to exactly 16 [4]. We repeated the combined-family analysis on this enlarged space. Restricting to Families 1+2, the kernel grows from dimension 2 to **dimension 3**: the new direction U_3 has a nonzero coefficient of the T -based basis element, confirming it is genuinely new, not a repackaging of U_1, U_2 . But U_3 also falls at x^{-22} in Family 3, exactly like U_1, U_2 .

To settle this definitively rather than checking individual candidates one at a time, we computed the combined kernel of *all three* families directly on the complete 16-dimensional space (31 constraint rows: 6 from Family 1, 7 from Family 2, 18 from Family 3, exact rational arithmetic throughout). The result: **rank exactly 16, kernel exactly $\{0\}$** . No combination of any of the sixteen known degree-11 building blocks satisfies $\leq s^2$ in all three trajectory families simultaneously. This was checked at two independent truncation orders for Family 3 ($N = 30, 45$), with identical coefficients across the entire range of orders used — the numerical stability required before trusting a high-order Laurent expansion.

	Family 1	Families 1+2	Families 1+2+3
Degree 9 (dim. 3, exact)	floor s^3	—	—
Degree 11, 15 known atoms	subspace dim. 9, $\leq s^2$	subspace dim. 2, $\leq s^2$	
Degree 11, + T atom (dim. 16)	—	subspace dim. 3, $\leq s^2$	kernel = $\{0\}$

TABLE 1. Summary of the degree-9/11 Regge-growth search. Each column to the right imposes the bound in one additional family simultaneously; the rightmost result is exact and definitive within the 16-dimensional space explored.

6. VERIFICATION METHODOLOGY

Beyond the gauge-reference bug of Remark 2.1 and the relabeling non-issue of Remark 4.1, two further real errors were caught and corrected:

- A first implementation of the fast Laurent-series backend (built after benchmarking showed sympy’s expression-tree overhead, not the underlying arithmetic, was the bottleneck: multiplying and truncating a series 2000 times took 106s in sympy vs. 1.6s with plain **Fraction** arithmetic) truncated the high-order end of each series after every multiplication, which is *incorrect* for Laurent series with negative orders: a discarded high-order term can re-enter the window of interest after multiplying by a sufficiently negative-order factor later. Caught by a 100-case randomized test against sympy; fixed by never truncating until the final read-out.

- Re-deriving Family 2 with the fast backend produced mismatches against the (already-trusted) sympy reference. Diagnosis: the chosen parameters left an uncanceled $\sqrt{3}$ in the *individual* (pre-symmetrization) polarizations, which only cancels after summing over the full S_5 orbit; the fast backend’s coefficient-conversion routine used Python’s `int()` on a sympy expression, which silently truncates an irrational like $\sqrt{3}$ to 1 instead of raising an error. Fixed two ways: the conversion routine now requires an exact integer numerator and denominator and raises an explicit error otherwise; and the Family 2 parameters were changed to values keeping $\sqrt{R(x)}$ exactly rational throughout, removing the irrationality at its source.

As with the companion note, this track record is informative but does not substitute for independent external verification.

7. DISCUSSION: WHAT THIS MEANS

Four structurally different attempts to find a degree-11 candidate compatible with CRG at $n = 5$ — three qualitatively distinct trajectory families, and an enlarged 16-dimensional search space including a genuinely new atom — all fail in the same qualitative way: candidates that survive some subset of the tests are killed by the next, and the combined constraint across everything tried is $\{0\}$. We think this pattern is more consistent with *the definition of the limit being wrong* than with *the search being incomplete*, though we cannot rule out the latter (see the caveats below).

This connects back to the two-problem split made at the start of this project (see the project notes): deriving CRG from first principles (via the chaos bound, as in [2]) was judged out of reach for this project, so we took CRG as a given conjecture and attacked only its consequences (“Problem B”). What this note shows is that, at $n \geq 5$, Problem B does not cleanly separate from Problem A the way it does at $n = 4$: one cannot even state the bound to be checked without first answering a scaled-down version of the holographic question — which kinematic regime is the one in which a causality/chaos-bound argument would actually constrain growth to s^2 at $n \geq 5$. That is a real open problem, and we do not believe it can be resolved by further search within the family of trajectories explored here.

Remark 7.1 (Honest caveats). *This conclusion depends on assumptions we have not independently verified: (i) that the CRG bound, if it generalizes to $n \geq 5$ at all, keeps the same exponent s^2 (rather than some n -dependent power); (ii) that the three families explored, despite being qualitatively distinct, are representative of “reasonable” generalizations of the Regge limit, rather than three correlated failures of a single underlying modeling mistake; (iii) that degree 11 (rather than 13, 15, ...) is the right place to look — the parity theorem of [4] rules out even degrees, but says nothing about whether higher odd degrees behave differently.*

Open problem. What is the correct, physically motivated generalization of the classical Regge limit to $n \geq 5$ -particle scattering, and does a version of the Classical Regge Growth bound hold there? We do not know the answer, and we do not think it can be settled by more computation of the kind in this note — it likely requires either a genuine extension of the holographic chaos-bound derivation of [2] to $n \geq 5$ correlators, or an independent flat-space causality argument that does not go through holography at all.

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